

PVL238: VLSI Architectures

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Course objective: The motive of this course is to inculcate the knowledge of the different processors; their architecture and organizational intricacies. For performance enhancement consideration is given to various instruction level and memory management techniques such as pipelining, parallelism, instruction scheduling, hierarchical memory management etc.

Introduction: Review of basic computer architecture, quantitative techniques in computer design, measuring and reporting performance. Overview of the features of Instruction set architectures of CISC, RISC and DSP processors. Processor organization and Architectural Overview. Design of RISC processor: Building datapath and Control, multicycle implementation.

Pipelining: Basic concepts, instruction and arithmetic pipeline, data hazards, control hazards, and structural hazards, techniques for handling hazards. Exception handling. Pipeline optimization techniques, dynamic instruction scheduling

Hierarchical memory technology: Inclusion, Coherence and locality properties; Cache memory organizations, Techniques for reducing cache misses; Virtual memory organization, mapping and management techniques, memory replacement policies.

Instruction-level parallelism: basic concepts, techniques for increasing ILP, superscalar, super- pipelined and VLIW processor architectures. Array and vector processors.

Multiprocessor architecture: taxonomy of parallel architectures. Centralized shared-memory architecture: synchronization, memory consistency, interconnection networks. Distributed shared-memory architecture

Course Learning Outcome (CLO):

The students will able to:

1. Describe the basics of different processors based on their architecture and organization.
2. Apply the concepts of pipelining to improvise the performance of RISC architecture.
3. Understand various memory organization and management techniques.
4. Acquire the knowledge of instruction-level parallelism and various advanced architectures.

Text Books:

1. Hennessy, J.L., Patterson, D.A, and Goldberg, D., Computer Architecture A Quantitative Approach, Pearson Education Asia (2006) 4th ed.
2. Leigh, E. and Ali, D.L., System Architecture: software and hardware concepts, South Wester Publishing Co. (2000).
3. Stallings, W., Computer Organization and Architecture: Designing for Performance, Prentice Hall (2003) 7th ed.

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